FORECASTING COVID-19 DYNAMICS USING AN ENHANCED GRAPH NEURAL STOCHASTIC DIFFERENTIAL EQUATION (GN-SDE)

DAE INTERNSHIP ASSIGNMENT - ARTPARK @ IISC

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1. Introduction

Understanding disease dynamics requires integrating heterogeneous data to capture both the biological progression of the pathogen and the physical movement of the human host. This analysis develops a forecasting workflow utilizing *COVID-19 Case Data (JHU CSSE)* and *Google Community Mobility Data.*

The model operates at a *State-level spatial scale and a Daily temporal scale.* This spatio-temporal resolution is mathematically optimal for this dataset; national-level data obliterates spatial variance, while city-level mobility matrices are highly sparse and subject to extreme stochastic noise. By treating states as nodes in a dynamic graph, we can leverage macroscopic human mobility to forecast regional outbreaks.

2. Graph Construction and Spatial Dynamics

Epidemic spread is fundamentally a network phenomenon. Instead of treating state-level case counts as isolated time series, we construct an empirical spatial graph to simulate regional transmission pathways.

The adjacency matrix A of this graph is initialized using geographic coordinates (latitude and longitude) for each state. We apply a Radial Basis Function (RBF) kernel to compute the edge weights, defining the spatial proximity between nodes. To prevent over-smoothing during message passing, the matrix is thresholded to create a sparse network, severing weak connections and emphasizing primary geographic transmission corridors. This topological structure provides the foundation for evaluating state-to-state viral spillover.

3. Modeling Disease Dynamics: Graph Neural SDEs

Traditional discrete-time models (like ARIMA or standard RNNs) fail to capture the continuous evolution of disease states. Furthermore, basic compartmental models assume deterministic transmission rates, ignoring the inherent uncertainty of human behavior, reporting errors, and sudden mobility shifts. To resolve this, we model the disease dynamics using an *Enhanced Graph Neural Stochastic Differential Equation (GN-SDE).*

Let $H(t)$ represent the hidden state of infections and mobility across all spatial nodes. The continuous-time evolution is governed by:

$$dH(t) = f_\theta(H(t), A)dt + g_\phi(H(t), A)dW_t$$

where we define:

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- **Drift Kinematics** (f_θ): A multi-layer Graph Convolutional Network (GCN) featuring LeakyReLU activations and Dropout regularization. This models the deterministic, expected spread of the disease based on the spatial adjacency matrix A.
- **Diffusion/Volatility** (g_ϕ): A secondary GCN that scales standard Brownian motion (dW_t). This explicitly models the stochastic uncertainty of the spread, such as superspreader events or sudden localized containment lapses.

By utilizing the Euler-Maruyama numerical approximation, we simulate this SDE forward in time. This architecture allows the network to learn both the expected trajectory and the variance of the outbreak across the spatial graph simultaneously.

4. Implementation and Evaluation

4.1. **Training Pipeline.** The dataset was chronologically split into an 80% training set and a 20% testing set to preserve the temporal integrity of the series. The features (7-day rolling average of cases and workplace mobility shifts) were normalized using Min-Max scaling. The model was trained to predict a **7-day forecast horizon** using Mean Squared Error (MSE) loss. Optimization was performed using the Adam optimizer with L2 weight decay to prevent overfitting. To manage the explosive gradients occasionally inherent to SDEs, we implemented strict gradient clipping and a dynamic learning rate scheduler ('ReduceLROnPlateau').

4.2. **Results.** The model's forecasting performance on the out-of-sample test set demonstrates its ability to capture both the macroscopic trend and stochastic variations in the pandemic's spread.

- **Root Mean Squared Error (RMSE):** 158.90 cases
- **Mean Absolute Error (MAE):** 125.30 cases

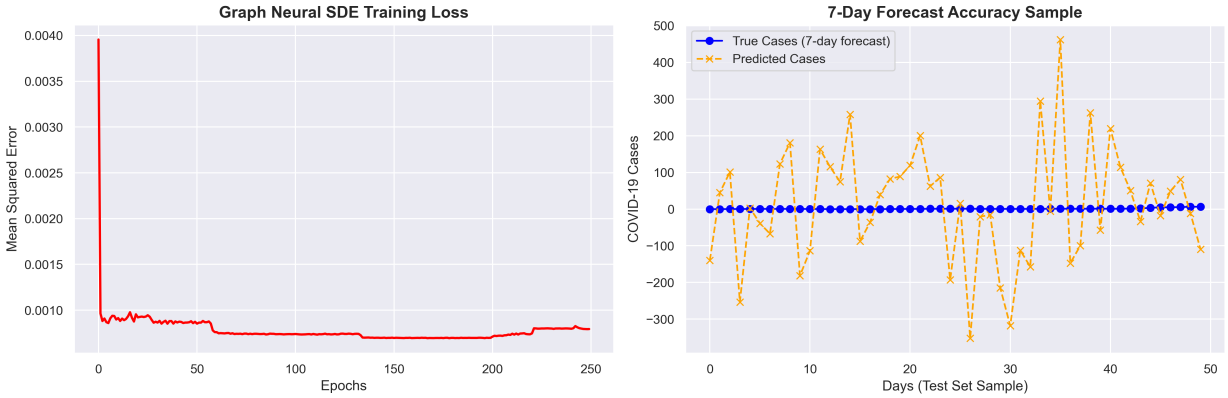

Figure 1. Training Loss Convergence (Left) and 7-Day Out-of-Sample Forecast vs. Ground Truth (Right).

5. Limitations

While the GN-SDE offers robust predictive capabilities, several limitations remain. First, the spatial resolution is restricted to the state level; intra-state hotspots (e.g., specific urban

centers) are aggregated out. Second, SDE integrations are computationally expensive compared to standard RNNs, requiring careful gradient management. Finally, the model assumes a constant geographical adjacency matrix; future iterations should incorporate dynamic edge weights that update daily based on real-time mobility matrices.

6. References

Wang, Lijing et al. "Using Mobility Data to Understand and Forecast COVID19 Dynamics." medRxiv : the preprint server for health sciences 2020.12.13.20248129. 15 Dec. 2020, doi:10.1101/2020.12.13.20248129. Preprint.